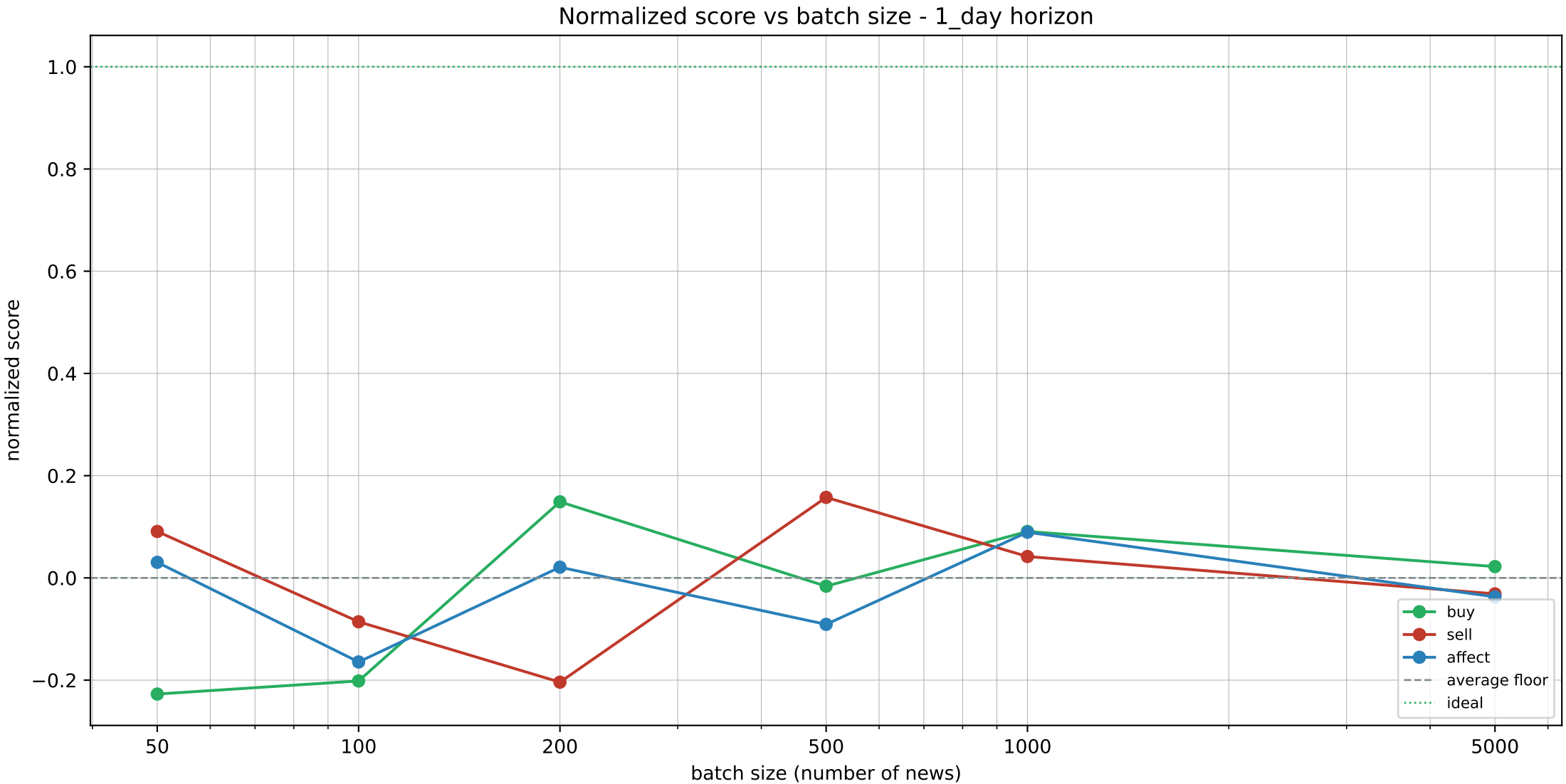
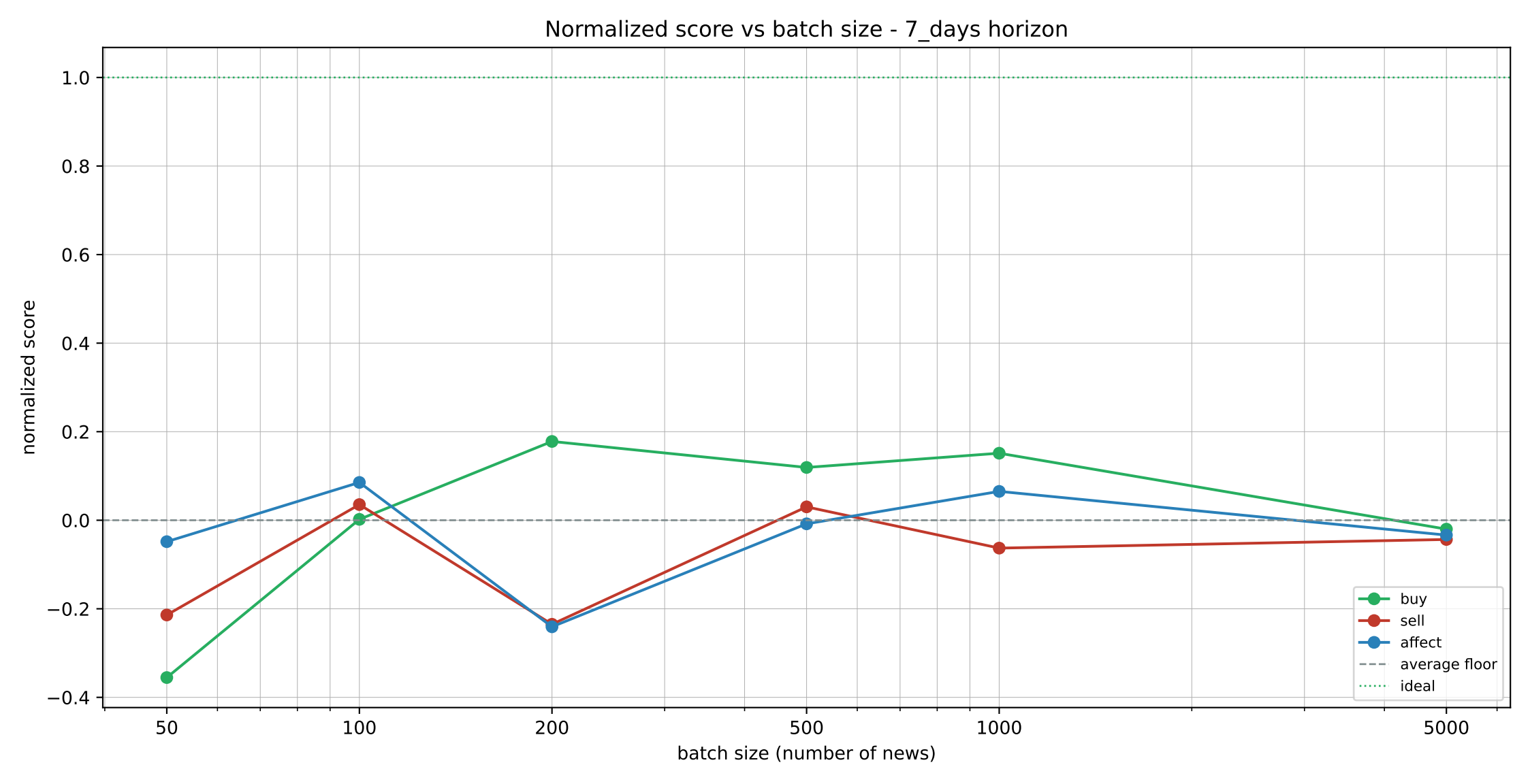
Filter evaluation report

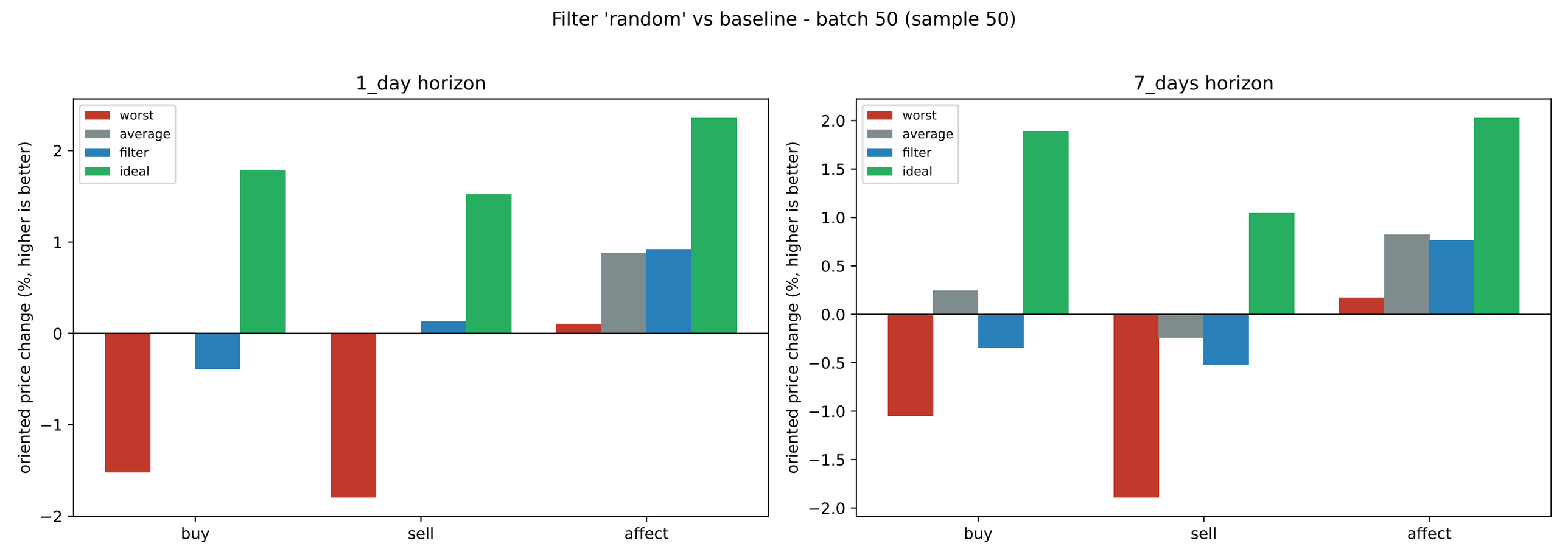
Filter: random
Dataset: sp100_2025-10-01_2026-01-01
Selection size (top_k): 10
Returns: market-adjusted (S&P 500 subtracted), standardized (z-score by 30-day volatility)
Random seed: 42
Batch sizes: 50, 100, 200, 500, 1000, 5000
Generated: 2026-06-30 20:31

Run overview

requested batch	sampled articles	filter time (ms)
50	50	0.07
100	100	0.05
200	200	0.05
500	500	0.04
1000	1000	0.05
5000	5000	0.06

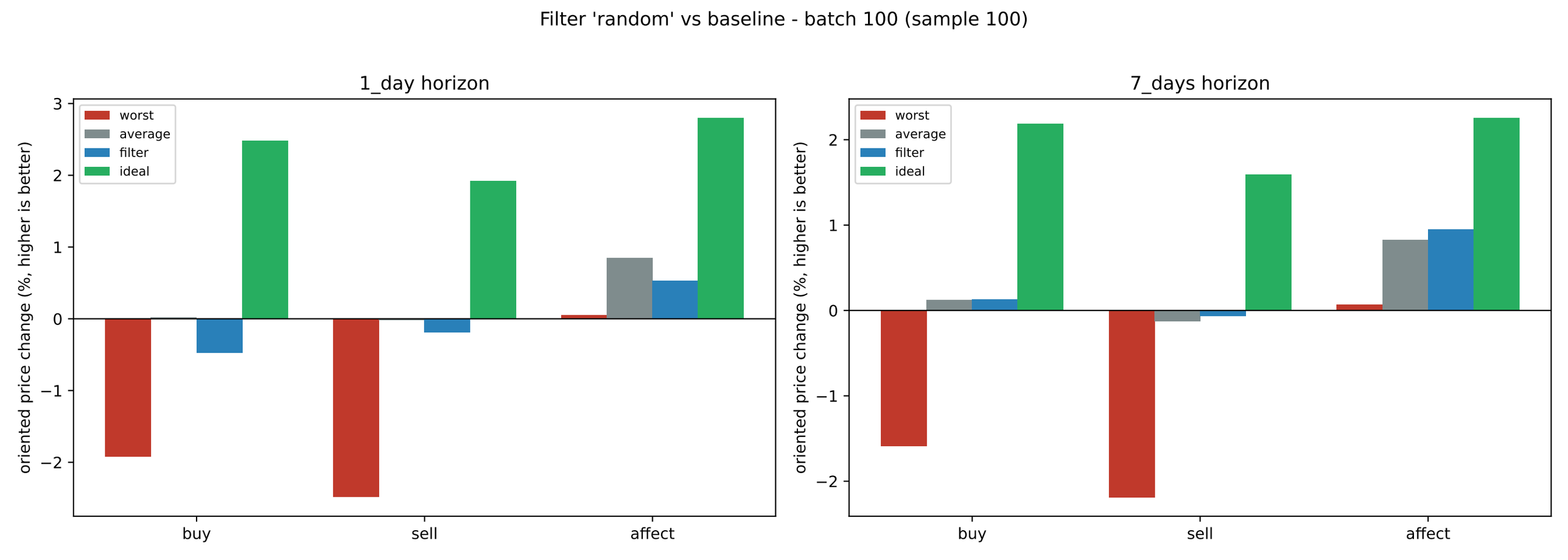

Normalized score vs batch size - 1_day horizon
normalized score
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
50
100
200
500
1000
5000
batch size (number of news)
buy
sell
affect
average floor
ideal


Normalized score vs batch size - 7_days horizon
normalized score
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
50
100
200
500
1000
5000
batch size (number of news)
buy
sell
affect
average floor
ideal


Filter 'random' vs baseline - batch 50 (sample 50)
1_day horizon
7_days horizon
worst
average
filter
ideal
oriented price change (%, higher is better)
buy
sell
affect

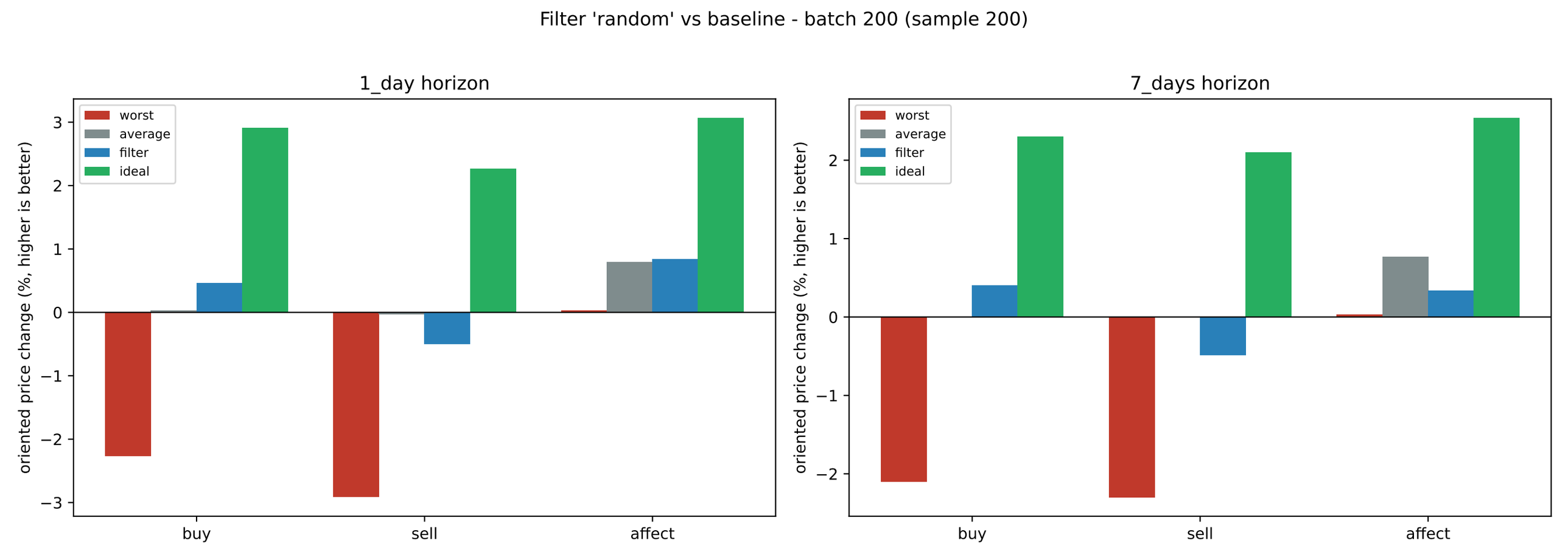
Oriented price change (%) - batch 50 (sample 50)

mode	horizon	filter	ideal	average	worst
buy	1_day	-0.394	+1.793	+0.011	-1.522
buy	7_days	-0.342	+1.889	+0.243	-1.047
sell	1_day	+0.128	+1.522	-0.011	-1.793
sell	7_days	-0.519	+1.047	-0.243	-1.889
affect	1_day	+0.922	+2.358	+0.876	+0.108
affect	7_days	+0.762	+2.028	+0.821	+0.170


Filter 'random' vs baseline - batch 100 (sample 100)
1_day horizon
7_days horizon
worst
average
filter
ideal
oriented price change (%, higher is better)
buy
sell
affect

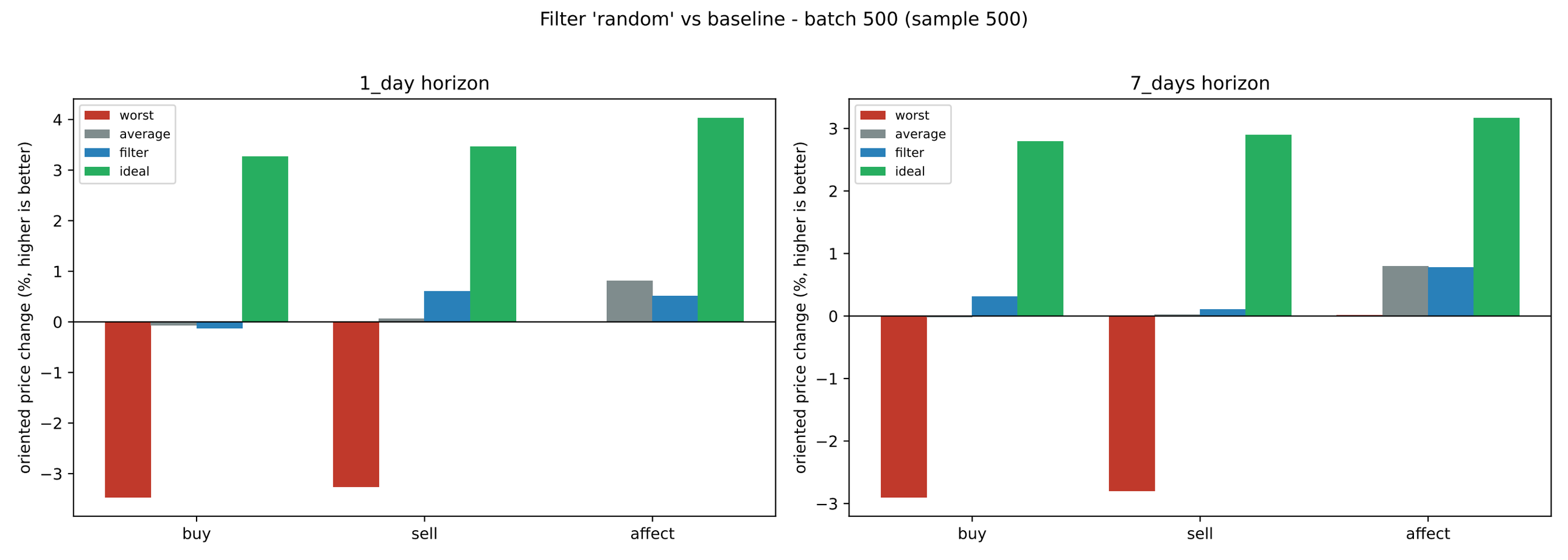
Oriented price change (%) - batch 100 (sample 100)

mode	horizon	filter	ideal	average	worst
buy	1_day	-0.477	+2.488	+0.021	-1.923
buy	7_days	+0.127	+2.188	+0.123	-1.591
sell	1_day	-0.187	+1.923	-0.021	-2.488
sell	7_days	-0.062	+1.591	-0.123	-2.188
affect	1_day	+0.529	+2.800	+0.850	+0.049
affect	7_days	+0.948	+2.255	+0.826	+0.066


Filter 'random' vs baseline - batch 200 (sample 200)
1_day horizon
7_days horizon
worst
average
filter
ideal
oriented price change (%, higher is better)
buy
sell
affect

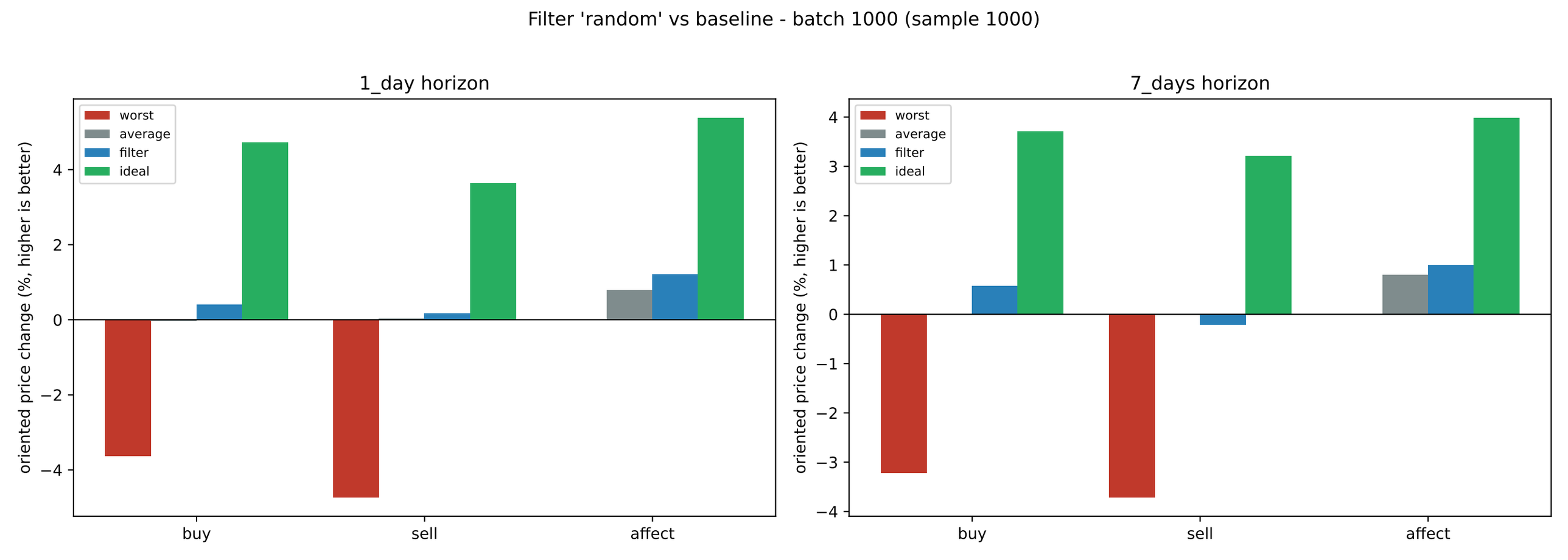
Oriented price change (%) - batch 200 (sample 200)

mode	horizon	filter	ideal	average	worst
buy	1_day	+0.462	+2.917	+0.032	-2.266
buy	7_days	+0.406	+2.299	-0.003	-2.100
sell	1_day	-0.501	+2.266	-0.032	-2.917
sell	7_days	-0.489	+2.100	+0.003	-2.299
affect	1_day	+0.842	+3.068	+0.794	+0.031
affect	7_days	+0.341	+2.539	+0.768	+0.032


Filter 'random' vs baseline - batch 500 (sample 500)
1_day horizon
7_days horizon
worst
average
filter
ideal
oriented price change (%, higher is better)
buy
sell
affect

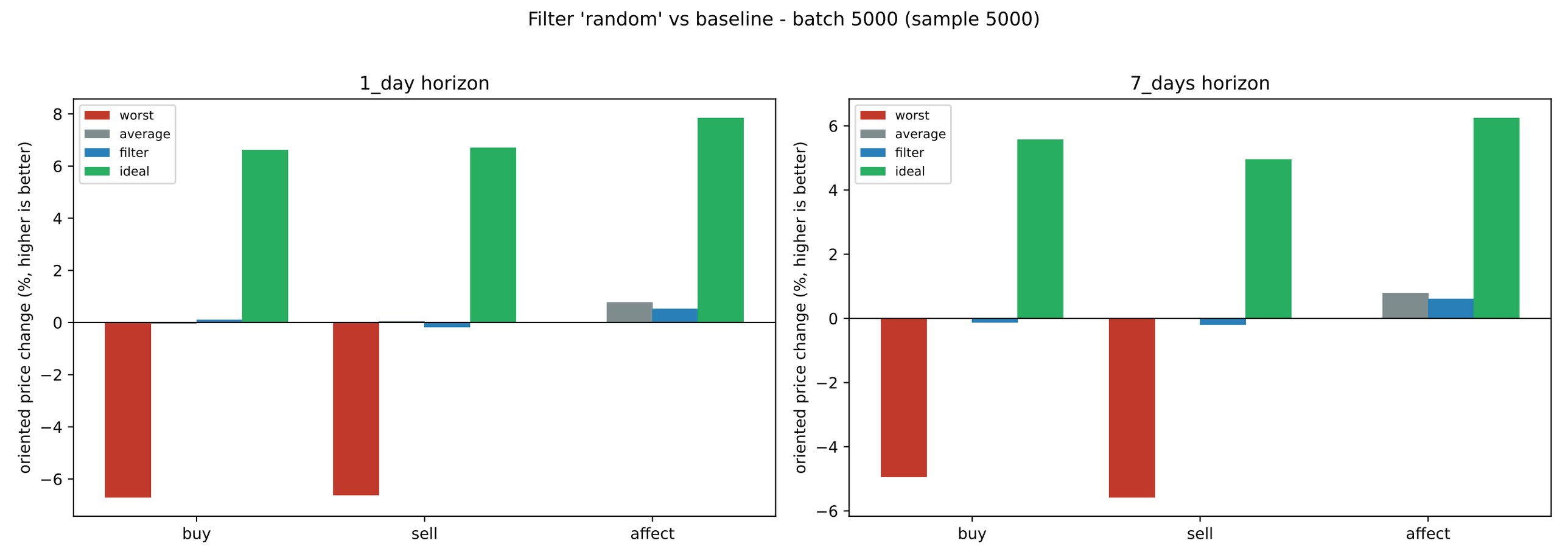
Oriented price change (%) - batch 500 (sample 500)

mode	horizon	filter	ideal	average	worst
buy	1_day	-0.121	+3.266	-0.067	-3.467
buy	7_days	+0.316	+2.799	-0.020	-2.900
sell	1_day	+0.602	+3.467	+0.067	-3.266
sell	7_days	+0.107	+2.900	+0.020	-2.799
affect	1_day	+0.518	+4.032	+0.811	+0.010
affect	7_days	+0.781	+3.169	+0.800	+0.016


Filter 'random' vs baseline - batch 1000 (sample 1000)
1_day horizon
7_days horizon
worst
average
filter
ideal
oriented price change (%, higher is better)
buy
sell
affect

Oriented price change (%) - batch 1000 (sample 1000)

mode	horizon	filter	ideal	average	worst
buy	1_day	+0.409	+4.729	-0.023	-3.633
buy	7_days	+0.571	+3.712	+0.010	-3.212
sell	1_day	+0.174	+3.633	+0.023	-4.729
sell	7_days	-0.213	+3.212	-0.010	-3.712
affect	1_day	+1.205	+5.378	+0.796	+0.005
affect	7_days	+1.003	+3.983	+0.795	+0.010


Filter 'random' vs baseline - batch 5000 (sample 5000)
1_day horizon
7_days horizon
worst
average
filter
ideal
oriented price change (%, higher is better)
buy
sell
affect

Oriented price change (%) - batch 5000 (sample 5000)

mode	horizon	filter	ideal	average	worst
buy	1_day	+0.101	+6.618	-0.047	-6.701
buy	7_days	-0.125	+5.575	-0.012	-4.945
sell	1_day	-0.160	+6.701	+0.047	-6.618
sell	7_days	-0.202	+4.945	+0.012	-5.575
affect	1_day	+0.520	+7.847	+0.783	+0.001
affect	7_days	+0.610	+6.247	+0.793	+0.002

Normalized score (0 = average, 1 = ideal)

mode	horizon	50	100	200	500	1000	5000
buy	1_day	-0.227	-0.202	0.149	-0.016	0.091	0.022
buy	7_days	-0.355	0.002	0.178	0.119	0.151	-0.020
sell	1_day	0.091	-0.086	-0.204	0.158	0.042	-0.031
sell	7_days	-0.214	0.035	-0.235	0.030	-0.063	-0.043
affect	1_day	0.030	-0.164	0.021	-0.091	0.089	-0.037
affect	7_days	-0.048	0.085	-0.241	-0.008	0.065	-0.034

Gap to ideal (lower is better)

mode	horizon	50	100	200	500	1000	5000
buy	1_day	+2.187	+2.964	+2.455	+3.387	+4.321	+6.518
buy	7_days	+2.232	+2.061	+1.893	+2.483	+3.141	+5.700
sell	1_day	+1.394	+2.110	+2.768	+2.864	+3.458	+6.861
sell	7_days	+1.566	+1.653	+2.589	+2.793	+3.425	+5.148
affect	1_day	+1.436	+2.270	+2.226	+3.513	+4.173	+7.327
affect	7_days	+1.266	+1.308	+2.198	+2.388	+2.980	+5.637

Lift over average (higher is better)

mode	horizon	50	100	200	500	1000	5000
buy	1_day	-0.405	-0.497	+0.429	-0.054	+0.432	+0.148
buy	7_days	-0.585	+0.004	+0.410	+0.336	+0.560	-0.112
sell	1_day	+0.139	-0.167	-0.469	+0.536	+0.151	-0.207
sell	7_days	-0.276	+0.061	-0.493	+0.087	-0.203	-0.214
affect	1_day	+0.045	-0.321	+0.048	-0.293	+0.409	-0.262
affect	7_days	-0.058	+0.122	-0.426	-0.019	+0.208	-0.183